

MAKERERE UNIVERSITY



**DIRECTORATE OF RESEARCH, INNOVATION
AND PARTNERSHIPS [DRIP]**

**RESEARCH AGENDA
2025/26 – 2029/30**



DIRECTORATE OF RESEARCH, INNOVATION
AND PARTNERSHIPS [DRIP]

RESEARCH AGENDA 2025/26 - 2029/30

ABBREVIATIONS AND ACRONYMS

AU	African Union
DRIP	Directorate of Research, Innovation and Partnerships
GoU	Government of Uganda
HSS	Health Systems Strengthening
KPI	Key Performance Indicators
MaK	Makerere University
MAKRIF	Makerere University Research and Innovation Fund
NARO	National Agricultural Research Organization
NDP	National Development Plan
NCDs	Non-communicable Diseases
NGOs	Non-Governmental Organization
PEAP	Poverty Eradication Action Plan
POC	Proof-of-Concept
SDGs	Sustainable Development Goals
STEM	Science Technology Engineering and Mathematics
UHC	Universal Health Coverage
UIRI	Uganda Industrial Research Institute
UVRI	Uganda Virus Research Institute
VR	Virtual Reality
4IR	Fourth Industrial Revolution

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FOREWORD



It is with great pleasure that I introduce the Makerere University Research Agenda 2025/26–2029/30, a cornerstone of our institution’s commitment to advancing knowledge, fostering innovation, and driving societal transformation. As a leading institution of higher learning in Africa, Makerere University continues to champion research excellence as a

critical pathway to addressing the most pressing challenges of our time. As a leading institution of higher learning in Africa, Makerere University continues to champion research excellence as a critical pathway to addressing the most pressing challenges of our time. This agenda represents our strategic vision for the next five years, focusing on harnessing interdisciplinary research, promoting innovation, and forging impactful partnerships to contribute to Uganda’s Vision 2040 and National Development Plan IV (NDPIV) as well as the broader global goals outlined in the United Nations Sustainable Development Goals (SDGs) and the African Union’s Agenda 2063.

The Makerere University Research Agenda 2025/26–2029/30 is a product of collaborative effort and extensive consultation with our academic community, industry stakeholders, policymakers, and development partners. It reflects our commitment to not only encouraging groundbreaking and interdisciplinary research that addresses critical societal needs but also translating research outputs into tangible solutions that benefit communities, industries, and the broader society. The Agenda is designed to guide our research endeavors towards thematic areas of national and global importance.



I invite you to engage with this agenda, whether you are a researcher, student, partner, or policymaker. Together, we can leverage the transformative power of research and innovation to create a better future for all.

On behalf of Makerere University, I extend my gratitude to all those who contributed to the development of this agenda, particularly the Directorate of Research, Innovation, and Partnerships for their leadership in this endeavor.

Let us work together to realize the vision of this agenda and reaffirm Makerere University's position as a global leader in research and innovation.

A handwritten signature in blue ink, belonging to Prof. Barnabas Nawangwe. The signature is stylized and fluid, with a long horizontal stroke extending to the right.

Prof. Barnabas Nawangwe [PhD]
VICE CHANCELLOR

MESSAGE



I am honored to share Makerere University's Research Agenda for 2025/26–2029/30, which reflects our unwavering commitment to academic excellence, innovation, and societal transformation. This agenda reaffirms the integral role of research in advancing the university's core mandate of teaching, research, and community engagement. As the Deputy Vice Chancellor for Academic Affairs,

I recognize the critical interdependence between research and teaching. Our academic programs thrive when informed by cutting-edge research that addresses global challenges and local priorities. This agenda provides a roadmap to foster a vibrant research culture, ensuring that our curriculum remains dynamic, relevant, and responsive to the rapidly evolving demands of society.

The outlined priorities in this research agenda align with Makerere University's strategic vision of becoming a thought leader and driver of sustainable development. By integrating innovative approaches, promoting interdisciplinary research, and enhancing partnerships, we aim to provide impactful solutions to pressing societal issues such as climate change, public health, governance, and technological advancement.

I extend my appreciation to the researchers, faculty, students, and partners who have contributed to the development of this agenda. Your dedication and ingenuity remain the cornerstone of our success.

I encourage all academic units to embrace this research agenda as a guide to strengthen the nexus between research, teaching, and community engagement.



Let us work together to transform lives, empower communities, and shape the future through impactful research.

A handwritten signature in blue ink, appearing to read 'M. Buyinza', written over a horizontal line.

Prof. Mukadasi Buyinza [PhD]

DEPUTY VICE CHANCELLOR, ACADEMIC AFFAIRS

PREFACE



It is with great pleasure that I present the Makerere University Agenda 2025/26–2029/30, a strategic document that underscores our commitment to advancing knowledge, fostering innovation, and addressing societal challenges through impactful research. As one of Africa’s leading institutions of higher learning, Makerere University has consistently

positioned itself as a hub of academic excellence and a catalyst for societal transformation. This research agenda is a testament to our dedication to contributing meaningfully to national, regional, and global development priorities, including Uganda's Vision 2040, the African Union’s Agenda 2063, and the United Nations Sustainable Development Goals (SDGs).

The Makerere University Research Agenda 2025/26 – 2029/30 was developed through a participatory process involving our esteemed faculty, students, external and stakeholders, community partners. It reflects the collective vision of leveraging interdisciplinary collaboration to address pressing issues such as climate change, public health, food security, technological innovation and social equity. This Agenda is anchored on three core pillars that align with the university’s mission namely advancing research excellence, fostering innovation and technology transfer as well as strengthening Community Engagement through inclusive and participatory approaches.

By focusing on interdisciplinary approaches and strategic partnerships, we aim to enhance the relevance and impact of Makerere University’s research contributions to Uganda and beyond.



As we embark on this journey, I call upon our academic community, industry partners, policymakers, and development organizations to join us in realizing the goals outlined in this Agenda. Together, we can harness the transformative power of research and innovation to create a sustainable and prosperous future for all.

Thank you for your continued support of Makerere University's research mission.

A handwritten signature in blue ink, appearing to read 'R. Wamala', with a large circular flourish at the end.

Assoc Prof. Robert Wamala [PhD]

**DIRECTORATE OF RESEARCH, INNOVATION, AND
PARTNERSHIPS**

1.0 Background

Established in 1922, Makerere University stands as a beacon of academic excellence on the African continent. Nestled in the heart of Kampala, Uganda, Makerere has evolved into a distinguished institution with a rich heritage of scholarly achievements, innovation, and community engagement. From its humble beginnings as a technical school, Makerere has grown into a comprehensive University renowned for its commitment to fostering intellectual curiosity and driving positive change. The University has been a crucible for shaping leaders, intellectuals, and change-makers, contributing significantly to the social, economic, and cultural development of Uganda and beyond.

Makerere University is implementing a 10-year Strategic Plan (2020/21-2030/31) to guide the delivery of her core functions while responding to the dynamic higher education context. The Strategy was aligned to a 5-year Strategic Plan [2020/21 - 2024/25] following the approval of the National Development Plan III 2020/21 - 2024/25. The Strategic Plan seeks to transform Makerere University into a research-led University, responding to national, regional, and global development challenges, as well contributing to global knowledge generation. To accomplish this goal, the University will leverage existing capacities to integrate knowledge generated through research, into teaching, learning, community engagement, and the world of work in order to maximize impact. It is envisaged that the profile of indigenous research will be increased during the strategic period, knowledge technology transfer promoted and Intellectual Property commercialized.

As we navigate the complexities of the 21st century, Makerere University remains unwavering in its commitment to research excellence. Our research agenda is deeply rooted in our core values of excellence, accountability, Professionalism, integrity as well as respect and inclusivity. It reflects our ambition to contribute meaningfully to global conversations while addressing the unique challenges faced by our country and region at large region.



Through this research agenda, Makerere University seeks to further position herself as a hub for transformative research, nurturing the next generation of scholars, and driving innovation that positively impacts society. As we embark on this journey, we invite our academic community, partners, and stakeholders to join us in advancing knowledge and shaping a future characterized by progress, inclusivity, and sustainable development.

1.1 Context Analysis

The Makerere University Research Agenda 2025/26 - 2029/30 was formulated in lieu of the national, regional and global context. Nationally, the plan is reflective of the range of current legal and policy frameworks including Vision 2040 and the Fourth National Development Plan (NDP IV)¹. On the other hand, the plan is globally anchored – among other frameworks - to the Sustainable Development Goals (SDGs) (2015) and Africa Agenda 2063. These contexts not only highlight the role of the University in national and global development but also influence the thinking behind the strategic direction. Table 1 highlights the extent of alignment and corresponding strategic direction for each of the policy frameworks.

¹ National Planning Authority (November 2024). Fourth National Development Plan (NDPIV) 2025/26–2029/30 [Draft]. Kampala, Uganda: National Planning Authority

Table 1: The Research Agenda in Light of the National, Regional and Global Context

Policy Framework	Analysis, Alignment and Strategic Direction
Sustainable Development Goals	The Sustainable Development Goals (SDGs) provide a comprehensive and globally recognized framework for addressing a wide range of economic, social, and environmental challenges. Universities play a crucial role in contributing to the achievement of the SDGs through research, education, and innovation. SDG 4 aims to "ensure inclusive and equitable quality education and promote lifelong learning opportunities for all". Universities contribute to the achievement of SDG 4 targets in the following way: (i) conduct research on education policies, methodologies, and best practices to enhance the quality and inclusivity of education systems; (ii) collaborate with government agencies, non-governmental organizations, and the private sector to establish partnerships that support educational initiatives; (iii) promote global citizenship education, fostering an understanding of global issues, intercultural competence, and a sense of responsibility towards sustainable development, (iv) offer continuing education programs, professional development courses, and other forms of lifelong learning, (v) expanding access to higher education, particularly for marginalized and underrepresented groups. For example, efforts to increase enrolment, provide scholarships, and create inclusive admission policies contribute to the SDG 4 target of ensuring equal access to quality tertiary education. By addressing these specific aspects of Goal 4, universities play a crucial role in advancing the broader agenda of sustainable development and ensuring that education systems contribute effectively to the well-being of individuals and communities.

Policy Framework	Analysis, Alignment and Strategic Direction
Fourth Industrial Revolution	Universities play a significant role in addressing various aspects of the Fourth Industrial Revolution (4IR), which is characterized by the integration of advanced technologies, data-driven decision-making, and the convergence of physical and digital systems. One specific aspect that universities often focus on is preparing students and society for the changing nature of work and employment in the 4IR. Among other aspects, universities ought to revise and/or develop curricula to incorporate emerging technologies and skills relevant to the 4IR. Universities need to emphasize the importance of digital literacy and technology skills across disciplines. Likewise, universities should form partnerships and collaborations with industry leaders to stay abreast of the latest technological advancements. By addressing these specific aspects of 4IR, Makerere's RA aims at producing a versatile and robust graduate that will effectively harness new local and global opportunities as they emerge.
Agenda 2063	Agenda 2063 is a strategic framework for the socio-economic transformation of Africa, developed by the African Union (AU). It outlines the continent's long-term vision for the next 50 years, with the goal of fostering inclusive growth, sustainable development, and integration. Universities can play a crucial role in addressing various aspects of Agenda 2063. One specific aspect that universities often focus on is human capital development and education. Nevertheless, universities play a crucial role of contributing to Agenda 2063 through: (i) providing quality education and skills development. The framework articulates the Africa we want predicting that at least 70% of all high school graduates will go on to have tertiary education with 70% of them graduating in the sciences, technology and innovation programmes; (ii) engaging in research and

Agenda 2063	goals of the framework, (iii) fostering local and international collaboration and partnerships with institutions outside Africa to bring in expertise, resources, and diverse perspectives, and (v) contributing to capacity building for effective governance and leadership.
Policy Framework	Analysis, Alignment and Strategic Direction
Makerere University Strategic Plan	The University developed a 10-year Strategic Plan (2020 – 2030) to guide the delivery of her core functions. The Strategy was aligned to a 5-year Strategic Plan (2020/21 – 2024/25). In optimising her potential as a knowledge hub, the University envisions “to be a thought leader of knowledge generation for societal transformation and development”. This would be realised through provision of transformative and innovative teaching, learning, research, innovations and services responsive to dynamic national and global needs. The overarching goal is to have “a sustainable research-led and responsive University with a highly productive workforce churning out versatile graduates”. The key interventions to guide the realisation of this plan include enhanced access opportunities and student experiences, use of digital tools and partnerships with industry; enhanced graduate training, research, innovations and technology transfer. The research agenda: (i) operationalizes the Makerere University’s vision by identifying specific research themes, priorities, and objectives that advance the institution’s Strategic objectives; (ii) demonstrates initiatives to address the University’s priorities; (iii) presents areas for collaborative research, (iv) includes metrics for measuring success - progress of research initiatives provides tangible evidence of strategic plan implementation.

Policy Analysis, Alignment and Strategic Direction Framework

EAC 2050	The East African Community (EAC) Vision 2050 articulates that investment in human capital is critical for wealth creation and employment. An educated population is required for effective and efficient production, knowledge transfer and technological adaptation and innovation. Given the current education challenges, the focus of Vision 2050 will therefore be on improving access to education and completion rates, particularly at higher levels. Universities can address the goals of education and human capital development outlined in the vision by offering programs that align with the region's economic needs, fostering innovation, and providing quality education. Further, universities contribute to research and innovation initiatives that support the development goals of the region. Likewise, universities have a central role of contribute of addressing healthcare challenges and promoting social development
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National Policy Frameworks	The Uganda Vision 2040 implemented through the National Development Plans identifies human capital development as one of the fundamentals that need to be strengthened to accelerate the country’s transformation and harnessing of the demographic dividend. The University is set to continue contributing to: (i) the proportion of qualified and skilled labour force transitioning into decent employment, (ii) research and innovation, contributing to the generation of knowledge, technological advancements, and solutions to address national challenges, (iii) national health goals by training healthcare professionals, conducting health research, and implementing programs that address public health challenges. The strategic interventions earmarked in the planning cycle will therefore address the employment and wealth creation structural bottlenecks identified in the NDPIV, including low labour productivity currently at 38 percent, low human development (HDI at 0.516) and fewer STEI graduates (2 out of 5). By 2030, Makerere University will have made tremendous achievement in turning around its student enrolment portfolio with graduate students projected to increase by at-least 30%. Graduate enrolment coupled with a robust research agenda will contribute to the development of national intellectual outputs and resources that meet the human capital needs of the country and region.
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1.1 Strategic Objectives of the Research Agenda

There are six strategic objectives that the University aims to achieve through its research activities. The objectives have been aligned with the University’s strategic plan and the overall mission, vision, and values of the University.

1. To create an enabling, harmonious, transparent and efficient environment for research and innovations



2. Cultivate strategic partnerships with other research institutions, industry partners, government agencies, and non-profit organizations to enhance research capabilities and address complex challenges collaboratively;
3. Enhance communication and outreach efforts to disseminate research findings to diverse audiences, including the public, policymakers, and industry stakeholders;
4. Facilitate the transfer of research findings and technologies from the University to industry, contributing to economic development and technological progress
5. Promote ethical research practices and responsible conduct of research, ensuring that the University's research activities align with high ethical standards;
6. Improve funding and resources to support research initiatives, infrastructure development, and the expansion of research programs

These objectives collectively guide the development and implementation of the University's research agenda, providing a framework for prioritizing resources, fostering a culture of research excellence, and maximizing the impact of research activities on both academic and societal fronts.

1.2 Directorate of Research, Innovation and Partnerships

In 1994 Makerere University Council approved the establishment of a School of Postgraduate Studies, which later became the School of Graduate Studies in 2010 and has now transformed into the Directorate of Research, Innovation and Partnerships (DRIP). The Directorate plays a key role of advancing the University's mission by overseeing and promoting research activities, fostering innovation, and establishing strategic partnerships. The specific functions and responsibilities of the DRIP include:

- 1. Promoting and Coordinating Research Activities:**
Fosters a vibrant research culture across different academic, administrative and research units in-line with the University's strategic goals.



2. **Facilitating Innovation and Technology Transfer:** Supports translation of research findings into practical applications and managing intellectual property rights to ensure that research outputs are effectively protected and utilized.
3. **Research Capacity Building:** The Directorate oversees and grows capacity in research, innovation, and outreach among faculty and students, providing training and resources to enhance their skills and effectiveness. This contributes to the overall research culture and productivity of the University.
4. **Enhancing Research Capacity and Infrastructure:** Works to develop the University's research capabilities by providing access to necessary facilities, resources, and training programs that support high-quality research.
5. **Securing Research Funding:** Assists researchers and graduate students in identifying potential funding opportunities for their projects.
6. **Overseeing Research Ethics and Integrity:** Ensures that all research activities comply with ethical standards and regulatory requirements, maintaining the integrity and credibility of the university's research endeavours.
7. **Establishing and Managing Partnerships:** Build and maintain collaborative relationships with other institutions to enhance research opportunities and resource sharing.
8. **Disseminating Research Outcomes:** Promotes the dissemination of research findings through publications, conferences, and other platforms.
9. **Promoting Community Engagement and Outreach:** Facilitates initiatives that apply research insights to community development, fostering a reciprocal relationship between the University and society.

By fulfilling these functions, the DRIP ensures that the University remains at the forefront of academic excellence, contributes to societal advancement through innovation, and builds meaningful collaborations that extend its impact beyond the campus. The DRIP builds and sustains most of its activities on collaboration and active partnerships with academic and administrative unit within the University as well as with external stakeholders and development partners.

2.0 Situation Analysis

A situation analysis involved a comprehensive examination of the current state, context, and challenges relevant to the University's research landscape. It provided a foundation for identifying research priorities, setting goals, and developing strategies for future research initiatives.

2.1 Performance of the University in the Previous Research Agenda (2013/14-2018/19)

The analysis of the performance of a predecessor research agenda served three major objectives namely, Evaluation of Goal Achievement, Publication and Dissemination Metrics as well as identifying the strengths and weaknesses of the predecessor research agenda. Understanding what worked well and areas that needed improvement to provide valuable insights for refining future research strategies.

2.1.1 Evaluation of Goal Achievement

The previous Research Agenda which commenced in 2013/14, presented several milestones. Key among these are highlighted by thematic area in Table 2.

Table 2: Performance of the Previous Research Agenda 2013/14 – 2018/19

Area	Achievement
Teaching and Learning	- Exceeded the enrolment target of 35,000 students (N = 38, 648); - Reviewed and harmonized academic programmes at both undergraduate and graduate levels; - Established frameworks for joint award of degrees through course offerings and supervision; - Increased number of courses on e-learning platform currently at 790; - Improved proportion of graduate student enrolment from 7% to 12%; - One satellite campus was established in Jinja offering several undergraduate programmes.

Area	Achievement
Research & Innovations	• Increased research output from 313 to over 1,000 publications per year. • Maintained the University rankings into the top 10 over the period under review. • Increased number of staff with PhDs from 420 to 797 out of 1,574 total academic staff over the plan period. • Increased number of collaborations both local, regional and international
Knowledge Transfer Partnerships	• Established Makerere University Intellectual Property Management Office • Reviewed the University Intellectual Property Management Policy; • Established Makerere University Incubation Centre. • Growing number of research networks and collaborations e.g., Mak-Bergen, Resilient African Network (RAN), Association of Research Universities in Africa (ARUA), Regional Universities Forum (RUFORUM) and African Centers of Excellence (ACE). • Over 250 Memoranda of Understanding (MoUs) signed with partners in North America (USA and Canada), East, West and Southern Africa; East and South Asia; Western and Central as well as Nordic Countries (Norway and Sweden) from 2020 to 2023. • Institutionalized field attachment programme for all undergraduate students.
Strengthen Research Management and Coordination	• Established the Grant Administration, Management and support Unit (GAMU); • Reviewed the University Research and Innovations Policy; • Reconstituted the Makerere University Intellectual Property Management Committee

Area	Achievement
	- Established the Grant Management Committee; - Established the Makerere University Data Dashboard – an institutional online repository for all University Data; - Established the Makerere University Research and Management system (RIMS) – an online system for tracking graduate supervision;
Improve Funding for Research and Innovations	- Established the Makerere University Research and Innovation Fund; - Increased opportunities for internal funding for PhD and pot-graduate research - Increased development partners' research funding from US\$ 12m in 2008 to US\$ 82m in 2020 and US\$ 90m in 2023.
Institutional Arrangements (Enablers)	- Reviewed the recruitment, promotion and retention policy for the academic staff. - Increased number of senior academic staff especially professors to 85, above the plan target of 78. - Increased library space by 8,000 m² and sitting capacity of over 4,000 user - Increased total seating capacity in library of 1:10 compared to the plan target of 1:5. - Total library resources increased to over 463,396 copies. - Increased internet Bandwidth from 68 Mbps to 1.3 Gbps by 2019, and upgraded the University storage system to 2 Terabytes. - Established the Mak Wireless Network (MAKAIR) hence increase in the Optical-fibre network coverage beyond the main campus i.e., CHS and MUARIK - Digitalized the University process: automation of business processes: tuition collection, library and research management systems, among others.

Area	Achievement
	- Increased the total built up space to over 20,000 m². - Instituted structures for resource diversification, specifically Makerere University Holding Company (Mak Holdings) and Makerere University Endowment Fund (MakeEF). - Secured various scholarship programmes to support bright but financially disadvantaged students to access higher education. - Increased females in leadership positions (20%) and academic ranks (from 2 to 13 professors).

2.1.2 Publication and Dissemination Metrics An examination of the publication output and dissemination activities associated with the predecessor research agenda was made. This included assessing the quantity, quality, and visibility of research outputs such as journal articles, conference papers, and reports. Table 3 show an analysis of published papers of staff at the University along the 6 research themes and 4 cross cutting areas.

Table 3: Publication output of staff along the University Research Agenda Themes [2022]

No	Research Agenda Theme	Journal Articles
1.	Education and Education Systems	71
2.	Agricultural (crop & Livestock) transformation, Food Food Security and Livelihoods	107
3.	Governance, Culture, Visual Arts, Social Justice, Communication and Sustainable development	217
4.	Health and Health Systems	532
5.	Natural Resources Management and Climate Change	197
6.	Science and Technology	110
	Cross-cutting Areas	
1.	Biotechnology	-

2.	Knowledge Translation	-
3.	Human Resource Development	-
4.	Gender	19
	Total	1,253

Note. Publication in peer reviewed journal as reported in the College Annual Reports

The finding from Table 3 show that majority of the publications in peer reviewed journals were in the health and health systems (n = 532), followed by Governance, Culture, Visual Arts, Social Justice, Communication and Sustainable development (n = 217) as well as Natural Resources Management and Climate Change (n = 197).

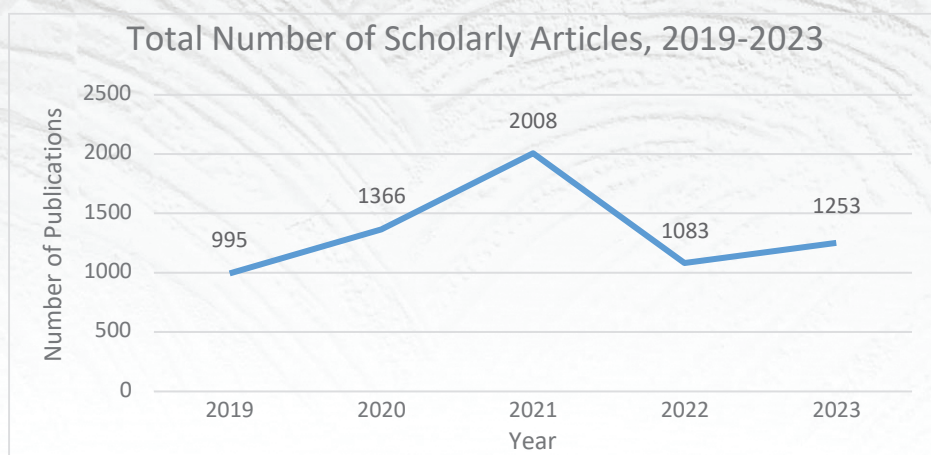


Figure 1: Publication output of staff at Makerere University in the past five years [2019 – 2023]

2.1.3 Government Support toward Research at the University

Makerere University received a first of its kind special funding from the Government of Uganda, to support Research and Innovations. The objective of fund was to support Research and Innovation initiatives that contribute to better delivery of National Development initiatives. The just concluded Financial Year (2022/23) is the fourth year of the fund, totalling 105.5 Billion UGX. Table 4 presents a summary of competitive grants awarded to researchers by Makerere University Research and Innovation Fund (MAKRIF). The presentation is made along the 14 research thematic areas of the fund. The priority areas were obtained from a comprehensive stakeholder consultation to develop a national Research and Innovation Agenda which was

conducted in 2019 by the Grant Management Committee (GMC).

Table 4: MAKRIF Competitive Grants awarded by Research

Thematic area	Financial Year				Total	%age
	2019/20	2020/21	2021/22	2022/23		
Sustainable health	162	79	68	31	340	39.3
Transforming the agricultural sector	56	44	20	12	132	15.3
Re-imagining Education	32	21	20	22	95	11.0
Catalysing business and enterprise	16	25	11	2	54	6.2
Water, sanitation and environmental sustainability	16	15	15	3	49	5.7
Harnessing the social sector to drive development	13	5	9	2	29	3.6
Energy and Minerals as drivers of rapid economic development	10	11	10	0	31	3.6
Leveraging public service and local administration for efficient services	11	9	6	2	28	3.2
Harnessing ICT to drive development	7	11	5	3	26	3.0
Harnessing tourism, wildlife and heritage to drive development	9	4	7	3	23	2.7
Sustainable Planning, finance and monitoring	1	3	8	5	17	2.0
Manufacturing, science and technology as tools to accelerate development	9	5	0	0	14	1.6
Strengthening law, governance, human rights and international cooperation	3	5	2	1	11	1.3
Defence and security: Achieving sustainable peace and stability	4	2	1	1	8	0.9
Total	349	239	190	87	865	100

Similar to the findings on number of publications in peer reviewed journals, the highest proportion of grants awarded were in health (39.3%). The finding from Table 4 show two other major areas of funding namely Transforming the agricultural sector (15.3%), and Re-imagining Education (11.0%).

2.1.3 Identification of Strengths, Weaknesses, Opportunities and Threat Analysis

It is important to identify the strengths and weaknesses of the predecessor research agenda to provides valuable insights for refining future research strategies. This involved the understanding of what worked well and areas that needed improvement.

Table 5: Makerere University's Strengths, Weaknesses, Opportunities and Threats

STRENGTHS		WEAKNESSES	
INTERNAL	- Established and effective governance structures - A pool of highly qualified and multidisciplinary human resource - University has hit a milestone of over 100 Graduates awarded Doctorates in each year - Continuous stream of bright students that can be nurtured into future academics - Makerere University brand - University estate on prime land that can be used to generate resources 7th best ranked university in Africa and among the best 500 globally in 2023		- Inadequate resourcing and maintenance of the existing Infrastructure - Underdeveloped infrastructure & information systems to promote e-Learning and modern pedagogies - Fragmented and individualised resource mobilisation initiatives - Recurring staff and student unrest - Weak mechanisms for conflict resolutions - Inadequate specialised training facilities for people with special needs - Inadequate linkage with the private sector - Policy shift in attracting talent at entry level (abolishing appointment of Teaching Assistant at entry level)
	OPPORTUNITIES		THREATS

EXTERNAL	• A supportive national & international policy environment • Globalisation and advances in ICT that support borderless learning • High and growing demand for university services (Teaching and Learning, Research & Innovation and KTTP) • Over 400,000 living and growing pool of alumni • A budding next generation of scholars/researchers and research leaders • A hub of centres of excellence for training, research and innovations in the region. • Legal provision as a National Repository for research works (publications).	• Absence of Higher Education Policy & Strategic Plan resulting into diversified and undifferentiated higher education system in the region • Increasing number of higher education institutions or universities established in the country • Rapidly changing technological advances. • Multiple discipline specific research centres competing for the same resource; • Unfavourable economic environment (small private sector) • Negative publicity • Limited number of technical personnel/training hubs for technicians. • Regulatory framework (the delayed review of the Universities and Other Tertiary Institutions Act, politics in the elective top management positions, etc.) • Culture, attitude and intellectual laziness - students reluctant to read and comprehend
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Source: Adapted from the Makerere-University-Strategic Plan 2020/21-2024/25

2.4. Emerging Issues from the Situational Analysis

From the situational analysis, there are a several opportunities that need to be harnessed by the University to swiftly adapt to the new changes and accelerate the delivery of teaching and research. The emerging issues from the situational analysis are presented in the following key areas:

1. The changing landscape in the delivery of higher education coupled with the digital revolution has brought about innovations aimed at transforming the student experience through a flexible and integrated learning environment.



2. The need to re-design the Institution's research and innovation ecosystem to create impact that enables the translation of research into tangible benefits to society.
3. Building strategic partnerships with a broad range of local and global networks is crucial to transformation of the Institution's research into commercial products.
4. A well governed institution is critical for professionally responsive, inclusive and efficient University operations.
5. Attracting, supporting and retaining a diverse and inclusive community of high achieving staff will be a key strategic enabler.
6. Infrastructure that matches the 21st Century University through the transformation of the university space and its efficient utilisation for an enhanced student experiences.
7. Financial sustainability through diversifying the university's income streams and managing its resources to establish a sustainable financial base.

3.0 Development of a Successor Makerere University Research Agenda

The process of developing a successor Makerere University Research Agenda began in June 2023 with Top Management meeting of the DRGT held on June 12th 2023. The meeting considered and approved the process and roadmap for the development of a successor Makerere University Research Agenda. Subsequently, an analysis of the performance of the predecessor University research agenda 2013-2018 was initiated and conducted in July 2023. The predecessor research agenda was approved by Senate in December, 9th 2013 with six (6) research priority areas namely (i) Health and Health Systems, (ii) Agricultural (crop & Livestock) transformation Food Security and Livelihoods, (iii) Natural Resources Management and Climate Change, (iv) Education and Education Systems; (v) Governance, Culture, Visual Arts, Social Justice, Communication and Sustainable development; (vi) Science and Technology. Further, the research agenda had four cross-cutting areas i.e. Gender, Quality Assurance, ICT and Biotechnology. The analysis served the purpose of assessing the extent to which the goals and objectives set in the predecessor research agenda were achieved. Further, the analysis helped to identify the



strengths and weaknesses of the predecessor research agenda so as to determine which areas were particularly successful and where there were challenges or shortcomings. The finding of the analysis provided a basis for the institution to learn from past experiences, optimize her research strategies, and also set the stage for a more impactful and successful future research agenda.

The second stage involved accessing all official documents, reports, and publications related to the University's research agenda. In other word, the successor Research Agenda was also guided by the following national and global aspirations as well as development frameworks:

1. **Frameworks:** *The review included, however was not limited to the following policy framework: National Development Plan III, Vision 2040, Sustainable Development Goals (SDGs), and Agenda 2063. The purpose of the review was to align the research agenda with national development plans, sector-specific strategies and global development agendas.*
2. **Institutional Documentation.** *Makerere University Strategic Plan 2020/21 – 2024/25, University Research Agenda 2013-2018, MaKRIF Research Agenda as well as College Annual Reports, Strategic Plans and Research Agendas.*
3. **Alignment with Institutional Mission and Goals.** *The alignment was undertaken to ensure that the proposed research agenda aligns with the overall mission, vision, and strategic goals of the university. Consider how research activities can contribute to the institution's reputation and impact.*
4. **Engagements on Research Agenda.** *It is also important to note that the review benefited tremendously from reports of research consultative workshops organized by MaKRIF and key government Ministries, Departments and Agencies including sectors and semi-autonomous government agencies and the other with the private-for-profit sector.*
5. **Benchmarking and Best Practices:** *We conducted a review of research agendas from peer institutions or those considered leaders in specific fields. The objective of the review was to identify est practices and successful models that align with the University's goals.*

It is important to note that process of developing a successor Makerere University Research Agenda was a collaborative one that involved input from various stakeholders, including University Top Management, faculty members, administrators, students, industry partners, and external experts. The relevancy of stakeholder consultation lies in the following key aspects: Informed Decision-Making, Understanding Diverse Perspectives, Building Ownership and Support, Enhancing Transparency and Accountability, Promoting Sustainable Development, Building Relationships and Trust; Compliance and Legal Requirements as well as conflict resolution. The developing a successor Makerere University Research Agenda benefited from related engagement of Research at the University as follows:

- 1. A stakeholder consultative engagement on Makerere University's Research priorities was held with Government Ministries, Departments and Agencies on August 2023. The engagement was collaboratively spearheaded by MaKRIF with attendance of over 50 participants;*
- 2. Subsequently, a stakeholder consultative workshop on Makerere University's Research priorities was held with Private Sector, NGO, CSO representatives; The engagement was collaboratively spearheaded by MaKRIF.*

On the other hand, the engagement with internal stakeholders of the University included, however was not limited to the following:

- 3. The Top Management meeting of DRGT was held on 7th December 2023 to consider a submission of the performance of the predecessor University research agenda 2013-2018, a desk review of relevant documentation as well as a record of the enabling environment that would guide the development and implementation of the successor Research Agenda;*
- 4. The Board of Research and Graduate Training - a committee with representation from all Colleges - met in January 2024 to harmonized the DRGT Top Management submission regarding the successor Makerere University Research Agenda. In particular, the Board considered the format of the successor Makerere University Research Agenda including research priority areas;*

5. *The Central Management Committee (CMC) of the University met in January 2024 to harmonize the Board of Research and Graduate Training submission regarding the successor Makerere University Research Agenda. In particular, the Committee considered the successor Makerere University Research Agenda including research priority areas;*
6. *Following the recommendations of the CMC, engagements were organized with each of the Colleges to disseminate the revised research agenda and also receive input from the units of the University. This activity was carried out between March 2024 to July 2024;*
7. *The Board of Research and Graduate Training considered the revised Makerere University Research Agenda on October 15th, 2024.*
8. *The revised Research Agenda was considered and approved by Senate on December 11th 2024.*

4.0 Scope

The Research Agenda has been formulated in lieu of the dynamic national, regional and global context. Nationally, the Agenda is reflective of the range of current legal and policy frameworks including Vision 2040 and the Fourth National Development Plan (NDP IV). Globally, the Agenda is anchored to the framework of the Sustainable Development Goals (SDG) (2015) and Africa Agenda 2063. These contexts not only highlight the role of the University in national and global development but also influence the thinking behind the research strategic direction.

Makerere University Research Agenda covers eight (8) priority research areas. The themes reflect the dedication of the University for reshaping the future through research excellence by aligning the institution's efforts to society needs fostering interdisciplinary collaborations and maintaining ethical standard.

5.0 Makerere Research Priority Areas 2025/6-2029/30

Following a documentation review and engagement with the stakeholders, the research thrusts to guide Makerere University for the next five years are:

- 1) Health and Health Systems
- 2) Agricultural Transformation, Food Security and Livelihoods
- 3) Natural Resources Management and Climate Change
- 4) Sustainable Urbanization and Housing
- 5) Education and Education Systems
- 6) Cultural Heritage and Preservation
- 7) Governance, Social Justice and Equity
- 8) Economic Growth, Business and Innovation
- 9) Science, Engineering and Technology

The sub-themes, key topics and development frameworks addressed by these research themes are as follows:

5.1 Theme I: Health and Health Systems

Health continues to be a priority area in national development and poverty alleviation. The role of health and the health system in national development is multifaceted, encompassing social, economic, and environmental dimensions. Prioritizing health as a fundamental component of national development strategies can lead to sustainable, inclusive, and resilient societies. This is also an area of Makerere's strength and is still relevant globally.

Sub-Research Theme	Key Topics	Development Agenda
Maternal, Child Health and Adolescent Health	Maternal and neonatal health interventions, Child growth and development studies, Family planning and reproductive health, improving prenatal care; Adolescent Health, Adolescent Sexuality, Childhood Blindness, Implantation Science, Neonatal and maternal morbidity and mortality	SDG 3, Universal Health Coverage, Health Systems Strengthening NDPIV
Healthcare Innovation and Technology	Adoption and impact of health technologies, Telemedicine and virtual care, Big data Analytics, Science, technology and health; Health informatics and data analytics	
Mental Health Services and Interventions	Interventions and support mechanisms for mental well-being Mental health policy analysis, Stigma reduction and mental health advocacy; School mental health, Workplace mental health, Alcohol and other substances of abuse and addiction.	
Infectious Disease, Zoonoses and Neglected Tropical Diseases (using one health approach)	Epidemiology and surveillance of infectious diseases including HIV, TB, Malaria, Micro-biology, Antimicrobial resistance, Post-operative infections, Emerging pandemics and Neglected tropical diseases; vaccination and immunization programs, vaccine hesitancy, zoonotic diseases, one-health	
Healthcare Quality Improvement	Continuous quality improvement in healthcare, Measurement and evaluation of healthcare quality; Implementation of evidence-based practices, development of scientifically accredited indigenous	

	medicines; determinants of access to medical oxygen, Parish Development Model and other government program	
Non-Communicable Diseases	Non-communicable Diseases (NCD) continue to pose a significant global health burden. These include sickle cell disease, Epilepsy, Encephalitis, Oral Health, Eye Diseases, Visual impairment and vision rehabilitation, Geriatric research, Basic and clinical neuroscience, Trauma and emergency care, Rehabilitation science Research is focused on prevention, early detection, and management strategies for NCDs	
Law, Policy and Governance	Policy development and implementation, Health systems governance and regulation, Policy impact assessment	

5.2 Theme 2: Agricultural Transformation, Food Security and Livelihoods

Agriculture continues to occupy a vantage position in the socio-economics of Uganda. Agriculture commonly refers to crop growing; thus, the need to highlight both crop and livestock. It goes beyond agriculture to include land use, inputs, methods and outputs. Transformation' is a pertinent concept that embraces issues of integration of agriculture with other sectors of the economy (industry, trade, technology etc...). On the other hand, Livelihoods' is all embracing to include household income, quality of life and other services e.g. health, water, transport among others

Sub-Research Theme	Key Topics	Development Agenda
Innovative agri-food systems for industrialization and improved livelihoods	Technological Innovation and Digital Agriculture, Sustainable Agricultural Practices and Productivity Enhancement; Integrated Food Systems for Nutrition Security and Public Health, Agricultural Innovation Systems and Rural/Urban Development; Crop Science and Crop Improvement, Optimization of Bioprocessing Techniques, Advanced Technologies for Food Safety; Preservation and Nutrition; Market Dynamics and Value Chains in Agribusiness; Biotechnological interventions for pest and disease resistance	SDG 2, Comprehensive Africa Agriculture Development Programme, Zero Hunger Challenge NDPIV
Sustainable landscapes and human settlements	Agroecological Approaches to Sustainable Land Use Planning, Soil Health Management in Land Use Planning; Urban Land Use Planning for Sustainable Development, Multi-functional Land Use Planning; Community Engagement and Participatory Land Use Planning; Sustainable Agriculture and Agro ecology.	
Livestock Farming and Animal Health	Sustainable livestock management, Disease prevention and control in animals, innovative diagnostic technologies, pharmaceutical development and regulation, one health approaches	
Agricultural and environmental policy and planning	Decision Support Systems for Sustainable Resource Management, Poverty Alleviation Strategies through Land Reforms, Agricultural Marketing Strategies for Smallholder Empowerment; Agrarian	

	Institutions and Contract Farming Models; Governance, Policy, and Environmental Sustainability; Policy coherence for sustainable development, Institutional frameworks for agricultural development	
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5.3 Theme 3: Natural Resources Management and Climate Change

Research in natural resources management, urbanization, and climate change is essential for understanding the interconnectedness of ecosystems, human settlements, and environmental health. This theme encourages research on effective management of resources like water, forests, and minerals, as well as on strategies to develop cities that are sustainable and adaptive to changing climate conditions. By exploring these issues, Makerere University will contribute to finding solutions for sustainable resource use, minimizing ecological footprints, and fostering urban resilience amidst growing environmental challenges.

<i>Sub-Research Theme</i>	<i>Key Topics</i>	<i>Development Agenda</i>
Environmental sustainability, climate change and resilience	Sustainable Environment and Natural Resource Management, Green Energy Sources and Energy Optimization, Innovative Environmental Technologies, Innovative Environmental Technologies; Food Systems; Community Resilience and Natural Disaster Risk Reduction, Nature-Based Solutions and Ecosystem Resilience; Climate-resilient infrastructure development, human world life conflict; Land use practices, planning and management, zoning, land development, land tenure	SDGs 6,7,11,13, 14 & 15 United Nations Framework Convention on Climate Change,

Water Resource Management and Agricultural Sustainability	Sustainable water use and allocation, Integrated water resources management, Watershed management, Climate change impacts on water availability, wetland management; governance for sustaining agricultural production	2030 Agenda for Sustainable Development
Renewable Energy and Energy Efficiency	Development of renewable energy sources, Energy efficiency measures and technologies, Climate-friendly energy policies and practices, Access to clean and sustainable energy;	
Animal genetic resources improvement and bio-technology	Genetic selection, conservation, assistant reproductive technologies and bio-banking	
Ecosystem Services and Biodiversity Conservation	Biodiversity monitoring and management, Integration of biodiversity into farming systems, Sustainable land use planning; Weather Forecasting and Climate Modelling	

5.4 Theme 4: Sustainable Urbanization and Housing

Urbanization is one of the defining trends of the 21st century, with profound implications for economic development, social well-being, and environmental sustainability. The theme seeks to address these complexities by fostering interdisciplinary research and collaboration. It emphasizes the need to create inclusive, resilient, and environmentally sustainable urban environments that enhance the quality of life for all. This theme integrates diverse research areas, including urban planning, housing, transportation, energy systems, health, governance, and environmental sustainability. The University will aim at contributing to global efforts to achieve sustainable development, particularly the United Nations Sustainable Development Goal 11, which focuses on making cities and human settlements inclusive, safe, resilient, and sustainable.



Sub-Research Theme	Key Topics	Development Agenda
Urban Sustainability, smart cities and Climate Resilience	Sustainable urban planning, governance design and management, Climate-resilient infrastructure in urban areas, Urban climate change adaptation strategies, revenue mobilization in urban areas, urban agriculture, circularity Assessment	SDGs 6, 7, 9, 11 &13
Infrastructure Development	Urban infrastructure, transport infrastructure and management, energy infrastructure, water infrastructure and management	
Urban Planning and Design	Sustainable Urban Growth, Resilient Infrastructure, Mixed-Use Development	
Housing and Shelter	Affordable Housing Solutions, Urban Slum Upgrading, Sustainable Building Materials,	African Union's Agenda 2063
Urban Transportation and Mobility	Public Transit Systems, Non-Motorized Transport, Urban Traffic Management	NDPIV
Urban Governance and Policy	Participatory Urban Governance, Regulatory Frameworks, Equity and Inclusion	
Urban Environment and Sustainability	Green Urban Spaces, Urban Biodiversity, Climate Resilient Cities, Sustainable Water Management	
Urban Data and Technology	Urban Data Analytics, Geographic Information Systems (GIS), IoT and Smart Technologies	
Urban Energy Systems	Renewable Energy Integration, Energy Efficiency, Urban Electrification	



5.5 Theme 5: Education and Education Systems

Research on education and education systems is essential in responding to the diverse challenges facing education today—ranging from equity and inclusion to curriculum relevance in an increasingly digital world. This theme emphasizes exploring new pedagogical approaches, integrating technology into learning environments, and addressing the barriers that hinder effective education delivery. By investigating these aspects, Makerere University will contribute to developing robust education systems that prepare learners with the skills, knowledge, and critical thinking abilities necessary for the 21st century.

<i>Sub-Research Theme</i>	<i>Key Topics</i>	<i>Development Agenda</i>
Educational Technology and Digital Education	Digital Pedagogy, instruction/learning design, E-learning, future education technologies, open education, online education, mobile learning and Artificial intelligence in Education	SDG 4, Education 2030: Framework for Action,
Curriculum Development and Reform	Design and evaluation of educational curricula, Curriculum alignment with societal needs, Cross-disciplinary and integrated curricula	

Inclusive Education	Education for students with disabilities, Inclusive classroom practices, Policy and strategies for inclusive education, Accessible learning materials and technologies, refugee education	United Nations Girls' Education Initiative, NDPIV, Vision 2030
Early Childhood Education	Importance and impact of early childhood education, Curriculum and pedagogical approaches for young learners, Parental involvement in early childhood education	
STEM Education	Strategies for promoting STEM education, Gender equity in STEM fields, STEM curriculum development, Partnerships between education and industry	
Adult and Community Education	Adult and community education, Lifelong learning initiatives, Continuous professional development, Distance and continuing education, youth Education, citizenship/mindset change education, extra mural education;	
Student Well-being and Health Education	Mental health in educational settings, Well-being programs for students, Prevention and intervention strategies, role of education in promoting mental health, child Psychosocial support	
Higher Education	Management and governance in higher education, teaching and learning of pedagogy, research and innovation in higher education, TVET, post-graduate studies, business education, policy and planning in higher education, quality of higher education, graduate employability, law policy and governance in education	

Humanities and Language Education	Language planning and promotion in education, role and relevance of humanities and language education, mother-tongue education, language pedagogy	
Teacher Professional Development	Professionalism in teacher education, education management and planning, continuous professional development	

5.6 Theme 6: Cultural Heritage and Preservation

Research on cultural heritage and preservation is crucial in recognizing and valuing the diverse histories, traditions, and practices that enrich societies. This theme encompasses the study of indigenous knowledge, historical artefacts, languages, and intangible cultural expressions that define and connect communities. By exploring these areas, Makerere University will contribute to documenting, preserving, and revitalizing cultural heritage in ways that honor and sustain its relevance in contemporary life. Additionally, such research fosters a deeper understanding of how culture shapes social, economic, and environmental dimensions within a global context.

<i>Sub-Research Theme</i>	<i>Key Topics</i>	<i>Development Agenda</i>
Heritage Site Management and Planning	Developing sustainable management plans for cultural heritage sites, Integrating community engagement and tourism management into site planning, better care and animals' welfare, cultural diversity	SDG 11, 12
Digital Heritage and Virtual Reality	Utilizing digital technologies to create virtual reconstructions of heritage sites, Exploring the use of VR and augmented reality for cultural heritage education and preservation, augmented reality	UNESCO Convention for the Safeguarding of

Heritage Tourism and Economic Development	Assessing the economic impact of heritage tourism, Developing sustainable tourism models that contribute to local economies	the Intangible Cultural Heritage, Vision 2030 NDPIV
Climate Change and Cultural Heritage	Assessing the impact of climate change on cultural heritage sites, Developing strategies for mitigating climate-related risks to heritage preservation	
Public Engagement and Education	Designing educational programs to raise awareness about cultural heritage, Engaging the public in heritage preservation through outreach and events	
Indigenous knowledge systems	Language preservation and processing, documentary heritage, parigraphy and oral history	
Art Practices and Management	Cultural heritage preservation, artistic entrepreneurship, studio exploration and practices, material development, curating, exhibitions and art history	

5.7 Theme 7: Governance, Social Justice and Human Rights

Research on governance, social justice, and human rights addresses related to diversity, inclusion, human rights, and social justice, aiming to address inequalities and promote a fair and just society. This theme is dedicated to examining the structures, laws, and policies that influence social well-being and exploring solutions that foster inclusivity, transparency, and accountability in governance. Through such research, universities play an active role in identifying root causes of injustice and proposing evidence-based solutions that support social equity, democratic governance, and human dignity

Sub-Research Theme	Key Topics	Development Agenda
Governance	Governance structures and models, leadership and management, stakeholder engagement and participation, technology and innovation governance, globalization and internationalization, crisis management and resilience	SDGs 1, 3, 5, 10, & 16 Universal Declaration of Human Rights
Gender Justice	Studies on gender-based discrimination and inequality, Intersectionality and the impact of multiple identities, Policies and interventions for gender equity; Healthcare Access and Equity	Convention on the Elimination of All Forms of Discrimination Against Women;
Economic Justice	Investigating income inequality and wealth disparities, Access to economic opportunities and resources, Social mobility and economic empowerment, Impact of disparities of salaries in the public sector, measurement of productivity across sectors of the economy	NDP IV, Vision 2030
Law Reforms	Existing and emerging transnational security threats (money laundering, terrorism, illicit wildlife trade, racketeering, human trafficking, corruption, cybercrime, drug trafficking, arms trafficking), penal institutions, transitional justice, international criminal law, non-custodial alternatives.	WHO Global Strategy and Action Plan on Ageing and Health
Disability Justice	Investigating accessibility and accommodations for individuals with disabilities, Social inclusion and empowerment of people with disabilities, Disability rights advocacy	

Aging and Gerontology	Biomedical Aspects of Aging, Psychosocial Well-being and Mental Health in Later Life, Social Gerontology; Economic and Financial Aspects of Aging, Technology and Aging; Law, Policy and Governance	
Alternative Dispute Resolution	Mediation, arbitration, conciliation, negotiation traditional justice systems	
Human Rights	Human Rights and Governance, constitutionalism, security and peace; refugee, migration and forced displacement, regional and international relations	
Digital Justice and Access	Investigating the digital divide and disparities in technology access, Internet and information access for marginalized communities, Technological inclusion and digital literacy initiatives	
Media Representation and Cultural Justice	Research on stereotypes and misrepresentation in media, The role of media in shaping perceptions of social justice issues	

5.7 Theme 8: Economic Growth, Business and Innovation

Economic growth today is increasingly linked to innovation, which emerges from research that addresses pressing societal challenges, creates new industries, and revitalizes existing ones. By focusing on economic growth and innovation, the University not only contributes to expanding knowledge but also serve as incubators for groundbreaking solutions, new business models, and skilled workforce development. This theme highlights the essential role of Makerere University in providing an intellectual ecosystem where ideas are nurtured, and solutions are crafted to meet local and global demands.

Sub-Research Theme	Key Topics	Development Agenda
Catalyse business and enterprise	Growth of small and medium enterprises; access to finance and capital for small businesses; Innovation and Technology Transfer, Building Industry-University Collaborations for Enterprise Growth, Skills Development for a Business-Oriented Research Culture	SDGs 1,8,9,11, 12, 13 NDPIV
Public Policy, Governance, and Economic Development	Tax policy and compliance, regulatory effectiveness, public-private partnerships, anti-corruption measures, and policy impact assessment; strengthening the budget process	
Financial Inclusion and Access to Capital	Microcredit and microfinance, mobile banking, financial literacy, impact investing, and gender inclusivity in finance; tracking the efficiency of financial reforms;	
Works and Manufacturing	Innovations in waste management, Smart Materials and New Production Technologies; Manufacturing Process Optimization and Lean Production, Sustainable Manufacturing Business Models and Economic Impacts; workforce Development and Skills for Modern Manufacturing, Waste reduction strategies	
Health Economics and Economic Impacts of Public Health Crises	Healthcare financing, economic burden of diseases, health insurance systems, and socioeconomic determinants of health.	
Commercialization and Management	Pathways to Scale for University Research and Innovations; Technology Transfer and Intellectual Property (IP) Management;	

	Entrepreneurial Ecosystems and Start-Up Support; Public-Private Partnerships for Innovation Scaling; Funding Models for Commercialization and Scale-up; Knowledge and Technology Dissemination to Industry and Society	
Digital Transformation and Innovation in Business	E-commerce, digital marketing trends, fintech, digital currency adoption, and cybersecurity in business.	

5.8 Theme 9: Science, Engineering and Technology

Science and technology are at the forefront of transforming societies, driving innovation, and addressing some of the most pressing global challenges. As hubs of knowledge creation and dissemination, universities play an essential role in advancing research that pushes the boundaries of scientific understanding and technological progress. This theme covers foundation science and caters for both basic research and applied research. The theme includes research in areas of infrastructure, ICT, innovations, information science, technology and engineering.

Sub-Research Theme	Key Topics	Development Agenda
Applied Research and Technology Development	Developing new technologies and applications, Innovation in engineering and design, Translating scientific knowledge into practical solutions	SDG 9, 17 Technology Facilitation Mechanism, African Science, Technology, and Innovation Indicators, NDPIV, Vision 2030
Information Technology and Computer Science	Software development and programming, Artificial intelligence and machine learning, Cybersecurity and information systems, and Big Data Analytics; natural language processing, human computer interaction, small area estimation;	
Space Science and Exploration	Astrophysics and cosmology, Space exploration and missions, Satellite technology and space-based applications, Kriging, incorporating geospatial data in statistical analysis and dissemination	
Emerging Technologies	Exploration of cutting-edge technologies including robotics and autonomous systems, ethical and societal implications of emerging technologies, digital skills gap, smart technologies and cloud computing	
Information science	Information retrieval, meta-literacy, new generation library, data and knowledge sovereignty, generative Artificial Intelligence, open science and knowledge, information ethics	
Science Communication and Public Engagement	Communicating scientific discoveries to the public, Public understanding of science, Engagement with diverse audiences on scientific topics	



It is evident that the research thrusts are quite broad and this is intended to promote multi-disciplinarity in addressing research problems and also build research capacity in the identified areas. The University's resources both internally generated and those sourced from funding agencies and development partners will be directed towards these priority areas of focus. However, researchers are free to source funding to address their research interests that are outside of the University priority areas

6.0 Ethical Considerations

At Makerere University, we recognize that research is not only a pursuit of knowledge, but also a commitment to ethical conduct, integrity and social responsibility. The research agenda places a strong emphasis on upholding the highest standards of ethical and responsible research practices across all our academic disciplines. We firmly believe that the impact of research goes beyond scientific advancement, extending to the wellbeing of individuals, communities and the environment.

The ethical considerations comprise, however are not limited to the following:

- a) **Research Integrity and Accountability:** Integrity lies at the core of our research endeavours. The University is dedicated to maintaining the highest levels of research integrity, including honest reporting, transparent methodology, and accurate representation of research outcomes.
- b) **Informed Consent and Human Subjects Protection:** Research involving human subjects is conducted with the utmost respect for their autonomy, well-being, and rights. Our researchers adhere to rigorous protocols for obtaining informed consent, protecting sensitive information, and ensuring the privacy and confidentiality of participants.
- c) **Animal Welfare and Research:** When research involves animals, we adhere to the highest standards of animal welfare and ethical treatment. Our researchers follow established guidelines for humane care, ethical treatment, and responsible use of animals in research, contributing to both scientific progress and the ethical treatment of living beings.

- d) **Data Privacy and Security:** We recognize the importance of safeguarding data privacy and security. Our researchers implement robust measures to protect personal and sensitive information, ensuring compliance with applicable regulations and ethical principles. We are committed to using data responsibly and only for research purposes.
- e) **Ethics in Emerging Technologies:** As research explores new frontiers in technology and innovation, we remain vigilant about the ethical implications of emerging technologies. Our researchers are committed to anticipating and addressing ethical concerns related to areas such as artificial intelligence, biotechnology, and data analytics.
- f) **Social Responsibility and Community Engagement:** We view our research agenda as a means to foster positive social change. Our researchers endeavour to actively engage with communities, stakeholders, and policymakers to ensure that research outcomes are relevant, equitable, and contribute to the well-being of society.
- g) **Transparency and Open Science:** We promote transparency in research by encouraging open science practices, such as sharing data, methodologies, and research findings openly and collaboratively. This fosters accountability, reproducibility, and the advancement of knowledge across the academic community.
- h) **Responsible Innovation and Technology Transfer:** Innovation is guided by responsible practices that take into account the potential societal, environmental, and ethical implications of new technologies. We actively collaborate with industry partners to ensure that innovations arising from our research are deployed in ways that benefit society while minimizing harm.
- i) **Continuous Education and Training:** We are committed to providing our researchers with ongoing education and training in ethical and responsible research practices. Workshops, seminars, and resources are available to empower our researchers to make informed decisions that align with ethical principles.

Through our dedication to ethical and responsible research practices, Makerere University aims at becoming a beacon of integrity and a model of how research can contribute to a better world while upholding the highest ethical standards.

7.0 External Partnerships

At Makerere University, we recognize that impactful research often transcends the boundaries of academia. Our research agenda is deeply committed to fostering strong and collaborative partnerships with external organizations, industry leaders, government agencies, and non-profit entities. These partnerships enrich our research endeavours, promote knowledge exchange, and enhance the real-world applicability of our findings. We shall therefore consider the following as some of the External Partnership Practices:

- a) **Strategic Collaborations:** We actively seek out strategic collaborations that align with our research themes and priorities. By partnering with organizations that share our commitment to advancing knowledge and solving complex challenges, we amplify the impact of our research efforts and contribute to global progress.
- b) **Industry Engagement:** Our research agenda emphasizes engagement with industry partners to bridge the gap between academia and practical applications. Collaborations with industry leaders not only provide valuable insights into real-world problems but also facilitate technology transfer, innovation, and the development of solutions that benefit society.
- c) **Government and Policy Partnerships:** We recognize the importance of engaging with government agencies and policymakers to influence policy decisions based on rigorous research outcomes. Our partnerships in this realm are aimed at addressing pressing societal issues, informing evidence-based policies, and driving positive change on local, regional, and national levels.

- d) **Non-Profit and Community Alliances:** Collaboration with non-profit organizations and community groups enables us to channel our research efforts towards projects that directly impact underserved populations, address social inequalities, and contribute to community development. These alliances promote inclusive research practices that resonate with our commitment to social responsibility.
- e) **International Networks:** Our research agenda extends beyond geographical boundaries, encompassing international collaborations that facilitate cross-cultural learning and foster a global perspective. These networks of researchers, institutions, and organizations contribute to a rich exchange of ideas, methodologies, and best practices.
- f) **Research Funding and Sponsorship:** External partnerships often involve research funding and sponsorship from industry partners, government bodies, and foundations. We are dedicated to transparently managing these collaborations, ensuring that the funding received aligns with our research priorities and does not compromise our academic independence or ethical standards.
- g) **Knowledge Exchange and Technology Transfer:** Our partnerships extend beyond financial support, emphasizing knowledge exchange and technology transfer. Through joint workshops, seminars, and collaborative projects, we actively contribute to the dissemination of knowledge, skill development, and innovation across sectors.
- h) **Ethical Considerations:** We approach external partnerships with a commitment to ethical principles and research integrity. Our collaborations are built on mutual respect, shared values, and a dedication to responsible conduct in research and innovation.
- i) **Continuous Evaluation and Improvement:** Our University is dedicated to continuously evaluating the effectiveness of our external partnerships. We seek feedback from partners to improve collaboration processes, address challenges, and seize new opportunities for mutually beneficial engagement.

Through our proactive approach to external partnerships, Makerere University aims at creating a dynamic research ecosystem that thrives on collaboration, innovation, and meaningful impact. By forging these alliances, we elevate our research to new heights and contribute to the advancement of knowledge for the betterment of society.

8.0 Funding Strategies

To execute the Makerere University research agenda and drive innovation across diverse disciplines, a well-defined and robust funding strategy is imperative. Our goal is to secure sustainable funding from a variety of sources to support groundbreaking research initiatives, foster collaboration, and elevate the reputation of Makerere University as a leading research institution.

Our funding strategy encompasses the following key elements:

- a) **External Grants and Partnerships:** We will actively pursue external research grants from government agencies, industry partners, foundations, and other relevant funding organizations. Collaborative projects with external partners will be prioritized to leverage resources, share expertise, and address real-world challenges.
- b) **Internal Research Funds:** Makerere University is committed to nurturing internal innovation. We will allocate a portion of our budget to support seed grants, pilot projects, and early-stage research efforts. These funds will promote novel and high-risk research ideas that have the potential to lead to groundbreaking discoveries, positioning our institution at the forefront of innovation.
- c) **Corporate Partnerships and Industry Collaborations:** We recognize the value of industry partnerships in advancing research with practical applications. Collaborations with corporations and industry leaders will provide not only funding but also access to real-world challenges and resources. In addition, joint research projects, technology transfer agreements, and sponsored research initiatives will be pursued to foster mutually beneficial relationships.

- c) **International Funding Opportunities:** The global nature of research and innovation offers opportunities for funding from international sources. We will actively explore partnerships with international research organizations, governmental agencies, and international foundations to diversify our funding portfolio.
- d) **Government Initiatives and Programs:** We will actively participate in government-sponsored research programs, innovation funds, and initiatives that align with our research priorities. Collaborating with governmental agencies will enable us to access critical funding resources and contribute to national research agendas.

By implementing this comprehensive funding strategy, we are confident that we will secure the necessary resources to propel our research agenda forward. This strategy aligns with our mission to provide innovative teaching, learning, research and services responsive to National and Global needs; thus, positioning Makerere University as a global leader in transformative research

9.0 Faculty Development

A thriving research agenda is contingent upon a vibrant and engaged faculty body. At Makerere University, we recognize the critical importance of nurturing and empowering our faculty members to excel in their research pursuits and scholarly endeavors. The faculty development initiatives are designed to foster a culture of continuous learning, collaboration, and research excellence.

The University shall prioritize the following:

- a) **Research Mentorship Programs:** We will establish a comprehensive research mentorship program promoting pairing early-career faculty members with experienced researchers.
- b) **Grant Writing Workshops:** We shall organize regular workshops focused on grant writing to equip faculty members with the skills to secure external research funding. We shall source for experts in grant proposal development to lead these workshops, providing insights into successful strategies and proposal evaluation

- a) **Interdisciplinary Collaboration Platforms:** We believe that diverse perspectives drive innovative research. We will encourage interdisciplinary collaboration platforms as well as collaborations by faculty members from various departments in executing research projects.
- b) **Research Ethics and Integrity Training:** Upholding the highest standards of research ethics and integrity is paramount. We shall offer training sessions in ensuring that faculty members are well-versed in ethical considerations related to research design, data collection, and publication.
- c) **Support for Early-Career Researchers:** Early-career faculty members often require additional support. We shall offer targeted resources, workshops, and mentoring to assist faculty in building a strong foundation for their research careers.
- d) **International Collaboration Opportunities:** International collaboration enhances the scope and impact of research. We will facilitate faculty exchanges, joint research projects, and international partnerships to promote global research collaboration.

10.0 Graduate and Postgraduate Research

At Makerere University, we are deeply committed to nurturing the next generation of scholars and researchers. The graduate and postgraduate research programs are designed to provide an immersive and rigorous research experience that equips students with the skills, knowledge, and passion to contribute meaningfully to their fields. We believe that graduate and postgraduate research plays a pivotal role in advancing the University's research agenda and expanding the boundaries of knowledge.

- a) **Faculty-Student Mentorship:** The University shall foster a culture of close collaboration between faculty mentors and graduate/postgraduate students. This will be achieved in-part through encouraging research projects that integrate graduate students.
- b) **Interdisciplinary Research Opportunities:** We recognize the importance of interdisciplinary collaboration in solving complex challenges. The application and award of projects will encourage students in interdisciplinary research projects, fostering a holistic approach to problem-solving.
- c) **Research Symposiums and Conferences:** We shall organize regular research symposiums and conferences where graduate and postgraduate students can showcase their research findings. These platforms will enable students to receive feedback, exchange ideas, and connect with peers and experts.
- d) **Funding and Scholarships:** We are committed to supporting students in their research pursuits. We will offer competitive research scholarships and funding opportunities to attract the brightest minds. Financial support will alleviate barriers to research engagement and ensure diverse participation.

11.0 Publications and Knowledge Transfer Partnerships

Central to our research agenda is the commitment to disseminate knowledge and research findings to a global audience. Our goal is to contribute not only to the advancement of academic scholarship but also to the betterment of society through the widespread sharing of innovative ideas and discoveries. Our publications and knowledge dissemination strategy encompass the following key elements:

- a) **High-Impact Journals and Publications:** We will prioritize publishing research in high-impact peer-reviewed journals across disciplines, ensuring that our findings reach a wide and influential audience.
- b) **Open Access Initiatives:** We are committed to open access publishing to ensure that our research is accessible to a diverse and global audience without barriers. We will work with existing university structures to explore partnerships with open-access journals and repositories to increase the visibility of our research.
- c) **Research Highlights and Summaries:** To bridge the gap between academia and the general public, we will produce research highlights, summaries, and infographics that distill complex research findings into accessible formats. These materials will be shared on our website, social media platforms, and through press releases.
- d) **Public Lectures and Seminars:** We will promote public lectures, seminars, and webinars where researchers can present their findings and engage in discussions with a diverse audience. These events will foster knowledge exchange and stimulate intellectual dialogue.
- e) **Engaging with Policy-Makers:** We will actively engage with policy-makers, governmental agencies, and non-governmental organizations to ensure that our research informs policy and drives positive societal change.
- f) **Supporting Innovation, Business Incubation and Spin-offs:** Supporting innovation, incubation, and spin-offs is crucial for the University to implement her research agenda effectively. These initiatives facilitate the translation of academic research into practical applications, contributing to economic development and societal advancement. Key measures include establishing Incubation Centers, Proof-of-Concept (POC) Funding, promoting entrepreneurial culture, streamlining Intellectual Property (IP) policies and fostering industry partnerships.

Through our comprehensive approach to publications and knowledge dissemination, we will amplify the impact of our research by reaching a diverse audience ranging from fellow academics to practitioners, policymakers, and the general public. By sharing our discoveries, insights, and innovations, we will contribute to the advancement of knowledge and the betterment of society at large.

12.0 Research Infrastructure Development

A robust research infrastructure is critical for fostering innovation, collaboration, and academic excellence in universities. The university is committed to implementing the following strategies to enhance infrastructural development for research.

- a) **Strategic Planning and Prioritization:** Undertaking a comprehensive needs assessment to identify existing gaps and prioritize infrastructure investments based on research goals and academic strengths
- b) **Resource Mobilization:** In addition to actively seeking grants from government agencies, international organizations, and private foundations focused on research development, the university will dedicate a portion of the annual budget to research infrastructure
- c) **Leveraging Technology:** Invest in high-speed internet, cloud computing resources, and advanced software for data analysis. Secondly, developing robust digital library systems to provide researchers with access to global academic resources.
- d) **Public-Private Partnerships:** Partner with industries for funding and establishing research parks or innovation labs. In addition, facilitate technology transfer agreements that support infrastructure development in exchange for research outputs.
- e) **Infrastructure Sharing:** Partner with other universities or research institutions to share facilities and reduce costs. Further, the University would pursue options of making University infrastructure available to external researchers under specific agreements.

13.0 Monitoring and Evaluation

An effective research agenda requires continuous monitoring and evaluation to ensure that our goals are being met, resources are being optimally utilized, and the impact of our research initiatives is being realized. We are committed to implementing a rigorous monitoring and evaluation framework that allows us to track progress, make informed decisions, and adapt our strategies as needed.

Our approach to monitoring and evaluation includes the following key components:

- a) **Key Performance Indicators (KPIs):** We will define a set of measurable KPIs aligned with our research goals. These KPIs will serve as benchmarks for assessing the progress and impact of our research initiatives.
- b) **Data Collection and Analysis:** Robust data collection mechanisms will be established to gather quantitative and qualitative data related to research outputs, outcomes, and impact. Data analysis will provide insights into trends, successes, challenges, and areas for improvement.
- c) **Research Output Metrics:** We will track metrics such as the number of research publications, citations, collaborations, patents, and other indicators of scholarly output. These metrics will reflect the productivity and quality of our research contributions.
- d) **Stakeholder Feedback:** Regular feedback from faculty members, students, industry partners, and other stakeholders will be collected to gauge their perceptions of our research initiatives. Stakeholder input will guide adjustments and improvements to our research strategy.
- e) **Research Impact:** We will evaluate the impact of our research on academic, societal, and economic levels. This includes assessing the practical applications of research findings and their contributions to solving real-world problems.

- f) **Student Involvement and Success:** Monitoring the involvement of graduate and postgraduate students in research projects and evaluating their achievements will reflect the effectiveness of our student-centric research initiatives.
- g) **Periodic Reviews and Reports:** Regular reviews and reports will be conducted to assess the progress of our research agenda against established goals and benchmarks. These reviews will inform strategic decisions and highlight areas for improvement.

Through a systematic approach to monitoring and evaluation, we ensure that our research agenda remains aligned with our institution's mission, responds to emerging challenges, and maximizes its positive outcomes. By analyzing data, gathering feedback, and making informed adjustments, we will continuously enhance the quality and relevance of our research initiatives.

14.0 Governance and Leadership

The DRIP will: (i) oversee the overall implementation of the research agenda; (ii) coordinate with Colleges, institutes, and research centers to ensure alignment with the agenda; (iii) report progress to the University management and stakeholders. The Deans and Department Heads will Integrate research priorities into faculty and departmental plans as well as ensure faculty and students are engaged in the research agenda. The Board of Research and Graduate Training provide strategic direction and monitor progress of the Research Agenda.

15.0 Timeline for Implementation

This section presents key milestones, activities, responsible entities, and their corresponding durations, facilitating the systematic realization of the agenda's objectives within defined timeframes. By aligning resources, stakeholders, and processes, the timeline serves as a guiding tool for effectively executing the Research Agenda while adapting to emerging opportunities and challenges.

Phase	Activities	Year	Responsible Entity
Phase 1: Inception	Finalize the Agenda & identify resources	2025	DRIP
Phase 2: Rollout	Launch initiatives, capacity building	2025/26	DRIP/Colleges
Phase 3: Mid-Term	Conduct evaluations, refine priorities	2027/28	DRIP
Phase 4: Consolidation	Scale successful projects expand partnerships	2029/30	DRIP

16.0 Communication Strategy

The DRIP will disseminate the research agenda to internal and external audiences. Further, the Directorate will regularly update stakeholders through newsletters, websites, and workshops on progress of the Research Agenda.



Makerere University Research Agenda
2025/26-2029/30



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